

TrES-1b

R-1263

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_210506 · created 2026-07-31T12:58:39+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459340.8753 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1338 +/- 0.0065
Transit depth (Rp/Rs)^2	1.79 +/- 0.17 [%]
Orbital Inclination (inc)	89.92 +/- 0.69
Airmass coefficient 1 (a1)	1.252 +/- 0.011
Airmass coefficient 2 (a2)	-0.2142 +/- 0.0049
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	None
Optimal Aperture	4.12
Optimal Annulus	10.73
Transit Duration (day)	0.1036 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1338 ± 0.0065 vs 0.1358 (deviation 0.25σ)
Duration fit vs published	0.1036 d vs 0.1042 d (deviation 0.42σ)
Mid-time O–C	-5.3 min
Depth SNR	16.8
Residual scatter	0.88 %

Light curve

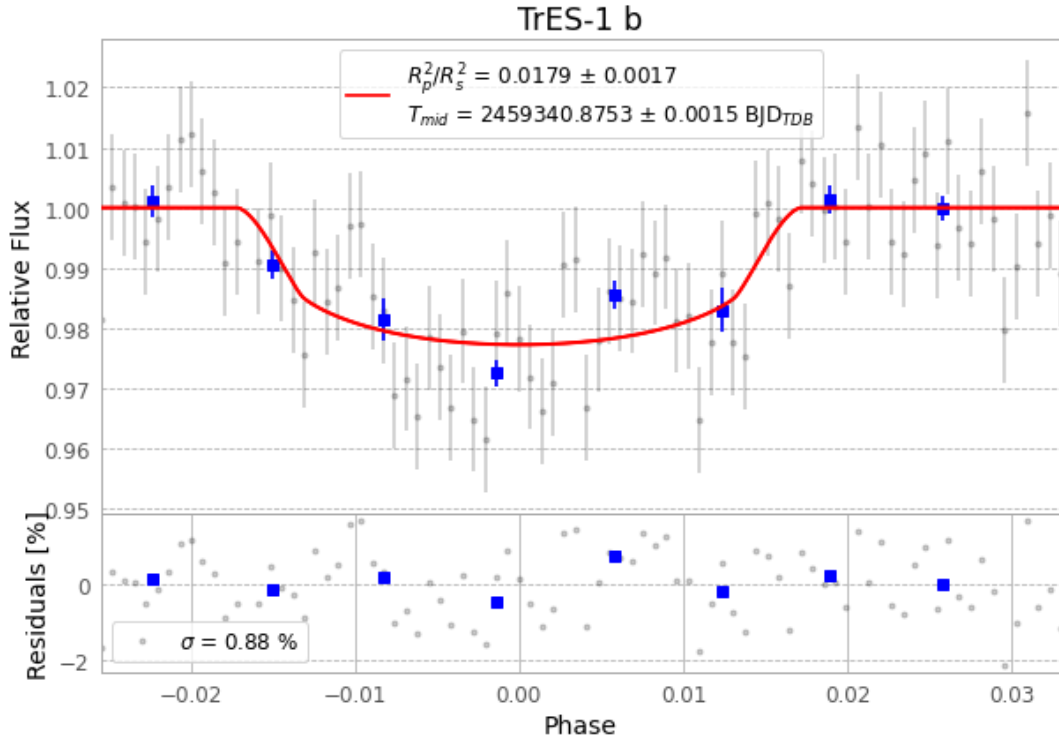


Figure 1 — FinalLightCurve_TrES-1 b_2021-05-06.png (SHA-256 c0964e6749298e04...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 235], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 292e78bd61205703...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), TRES-1210506070612.FITS → TRES-1210506112112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-210506.log (e0d89f394a1a...), FinalLightCurve_TrES-1 b_2021-05-06.png (c0964e674929...), FinalLightCurve_TrES-1 b_2021-05-06.csv (a3009400d3f4...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 12:58Z.